What Target-wise ABA Learning Does on Deterministic Causal Fixtures

Evidence from controlled probes H0–H7b

Working synthesis for discussion with Fabrizio

4 August 2026

Status and purpose. This is a supervisor-facing working document. It introduces the causal fixture, learner input, terminology, configurations, and experimental sequence before stating any cross-cutting findings. The controlled probes labelled H0–H7b are complete and analysed; the labels are internal shorthand only and each probe is described in the text by its scientific question and design. No cross-cutting conclusions are asserted yet. The six Findings headings and the Implications heading at the end are intentionally empty placeholders for subsequent drafting and review.

Abstract

The investigation asks what target-wise ABA Learning does when given only a finite binary table sampled from a known causal data-generating process. The principal process is a binary collider $A \to C \leftarrow B$ with independent stochastic roots and a deterministic AND child $C = A \wedge B$. The graph, mechanisms, population distribution, finite sample, learner-visible encoding, learned ABA framework, and evaluator-only reference are kept separate throughout.

The experimental sequence is recorded under labels H0–H7b for reference. It begins with a baseline comparison of the published Greedy and non-deterministic configurations across all targets on one frozen table (H0), then introduces one controlled question at a time: finite-sample support (H1); an irrelevant trailing covariate (H2); background-knowledge order with a leading distractor (H3); brave versus cautious learning (H4–H5); folding-token budget and grounded sample size (H6); and the `relto` versus `sechk` assumption-introduction options (H7). This document records what each probe was intended to reveal, its principal signal, and the procedural explanation supported by the generated deltas and Prolog traces.

Contents

1 Purpose, scope, and organising question

1.1 Research question

> For a finite binary table generated by independent stochastic roots and deterministic non-root mechanisms, and for each observed variable taken in turn as the learning target: what ABA framework does ABALearn emit under the configurations studied here, and how does that framework relate to the evaluator-only mechanism reference and to the information present in the population and in the sample?

The purpose is to describe how the learner transforms the supplied table into rules, assumptions, and contraries, and to classify each emitted theory against a target-dependent evaluator reference. On the principal AND collider $A \to C \leftarrow B$, only the deterministic child C has a desired learned rule over its observed parents:

```
c(A) :- a_val_1(A), b_val_1(A).
```

The roots A and B are generated by independent stochastic mechanisms. There is no underlying deterministic rule relating either root to the other observed variables, and none should be learned.

Relative to that target-dependent reference, the outcomes distinguished in this document are:

- for a deterministic child: a rule that coincides with the mechanism;
- for a deterministic child: a sample-adequate but non-minimal rule;
- for a deterministic child: a defeasible representation that differs from the mechanism string;
- for a root or child: a rule that holds on the finite sample because a population counterexample is missing from that sample;
- for a root: an assumption construction that satisfies the selected learning semantics while leaving the target underdetermined by its predictors;
- search termination under the configured procedure, including timeout or `completed_no_solution`, which for a stochastic root target may be the intended outcome under the reference above.

These classes separate procedural exit status, finite-sample adequacy, and mechanism alignment under the target-dependent reference.

1.2 Scope

The evidence comes from three small binary fixtures, fixed seed-42 samples, and the learner configurations described below. Learning is target-wise: one frozen table is held fixed, and each observed variable is selected in turn as the learning target. Changing the target changes the positive and negative examples and the background-knowledge encoding of the remaining columns; the underlying rows remain the same sample.

Only the deterministic child has an evaluator-desired deterministic rule over its observed parents. Root-target runs are diagnostics of the same pipeline when the target itself is generated stochastically: they show how finite support, learning semantics, and search budget shape the emitted theory or search outcome when no underlying deterministic rule from the other variables should be learned.

1.3 Object of study

The pipeline under study is target-wise ABA Learning over encoded tabular data. For each target it receives background knowledge and labelled examples only, and it emits an ABA framework of rules, assumptions, and contraries under a selected learning semantics and search configuration.

The evidence recorded here therefore concerns learned rule structure, search behaviour, and the relation of those outputs to the evaluator-only mechanism reference for the fixture under study.

2 Conceptual framework and notation

This section fixes the objects, ABA notions, target-wise encoding, and learner controls used throughout the experimental sequence.

2.1 Seven objects kept separate

The experimental reasoning distinguishes the following objects:

1. a generating DAG G;
2. structural mechanisms for the variables in G;
3. the exact observational population distribution P^*;
4. a finite IID sample $\mathcal{D}_n \sim P^*$;
5. target-specific learner input derived from $\mathcal{D}_n$;
6. the learned ABA framework and its emitted delta (the learned additions relative to the input background knowledge); and
7. evaluator-side judgements about sample adequacy, mechanism alignment, and minimality relative to the target-dependent reference.

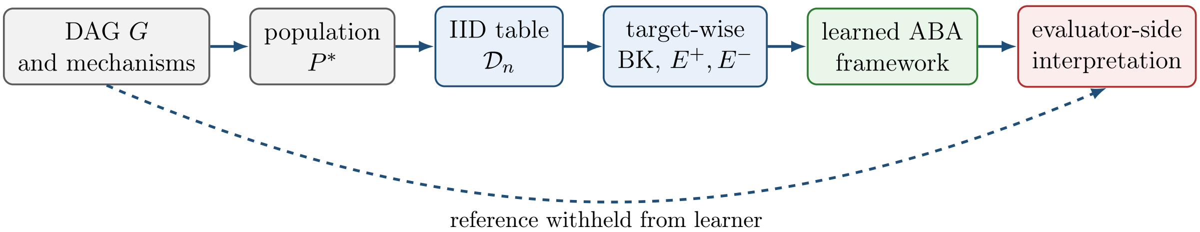

Figure 1: Experimental separation. Only the finite table and its target-wise encoding enter the learner. The generating graph and mechanism reference are reserved for evaluation.

Ordinary faithfulness is a property of the pair (P^*, G). Finite-sample adequacy is a property of a learned framework relative to $\mathcal{D}_n$ under the selected learning semantics. A parent literal appearing in a learned body is a syntactic fact about the emitted delta; any judgement that the delta aligns with the mechanism is an evaluator-side reading.

2.2 ABA framework

An ABA framework is a tuple

$$\mathcal{F} = \langle \mathcal{L}, \mathcal{R}, \mathcal{A}, \bar{\cdot} \rangle,$$

where $\mathcal{L}$ is a language, $\mathcal{R}$ is a set of rules, $\mathcal{A} \subseteq \mathcal{L}$ is a set of assumptions, and $\bar{\cdot}$ maps each assumption $\alpha \in \mathcal{A}$ to its contrary $\overline{\alpha} \in \mathcal{L}$.

A rule has the form

$$h \leftarrow b_1, \ldots, b_m.$$

If $m = 0$, it is a fact. Write $\Delta \vdash_{\mathcal{R}} s$ when a claim $s \in \mathcal{L}$ is derivable from a set of assumptions $\Delta \subseteq \mathcal{A}$ using the rules in $\mathcal{R}$.

Attacks are generated by contraries: Δ attacks an assumption $\alpha \in \mathcal{A}$ when $\Delta \vdash_{\mathcal{R}} \overline{\alpha}$. A set $\Delta \subseteq \mathcal{A}$ is *conflict-free* if it attacks none of its own members. It is *closed* if whenever $\Delta \vdash_{\mathcal{R}} \alpha$ for some $\alpha \in \mathcal{A}$, then $\alpha \in \Delta$.

A set $\Delta \subseteq \mathcal{A}$ is a *stable extension* of $\mathcal{F}$ if it is closed, conflict-free, and attacks every assumption in $\mathcal{A} \setminus \Delta$. Write $\mathrm{Stab}(\mathcal{F})$ for the collection of all stable extensions of $\mathcal{F}$.

A claim $s \in \mathcal{L}$ is accepted with respect to a stable extension Δ if there exists $\Delta' \subseteq \Delta$ such that $\Delta' \vdash_{\mathcal{R}} s$; write $\mathcal{F} \models_{\Delta} s$ when this holds. Write $\mathcal{F} \models_{\text{caut}} s$ when s is accepted with respect to every stable extension of $\mathcal{F}$. A learned framework may therefore combine ordinary rules with explicit exceptional conditions expressed by assumptions and contraries.

2.3 ABA Learning task

An ABA Learning task begins with background knowledge and two finite sets of examples:

$$E^+ = \{e_1^+, \dots, e_p^+\}, \qquad E^- = \{e_1^-, \dots, e_q^-\}.$$

The learner applies symbolic transformations to construct a framework $\mathcal{F}'$ under which the examples satisfy the selected consequence condition. The learned object is itself an ABA framework.

The transformations relevant here are:

- **Rote learning.** Add example-specific rules from the positive examples.
- **Folding.** Replace example-specific conditions with reusable BK predicates, producing intensional rules.
- **Subsumption.** Remove a specialised rule when an appropriate more general rule already exists.
- **Assumption introduction.** Make an over-general rule defeasible by adding an assumption, then learn rules for its contrary to represent exceptions.

2.4 Exact-value target-wise encoding

In this investigation the learning task is instantiated from a tabular sample. For each row identifier i, every non-target variable is represented by exactly one value predicate. When c is the target, for example, the BK contains facts such as

```
a_val_1(9).
b_val_0(9).
```

and row 9 contributes either $c(9) \in E^+$ or $c(9) \in E^-$, according to its target value. For a binary target Y,

$$E_Y^+ = \{Y(i) : Y_i = 1\}, \qquad E_Y^- = \{Y(i) : Y_i = 0\}.$$

All non-target columns are available as BK predictors in their serialised order. The encoding interface is the same for every target. The evaluator-side desired object still depends on the target's generating role: only a deterministic child has a desired parent mechanism rule.

2.5 Brave and cautious learning

Brave and cautious learning are the two consequence conditions under which a learned framework $\mathcal{F}'$ may be required to cover E^+ and exclude E^-.

Brave learning requires an existential witness: there must exist some $\Delta \in \mathrm{Stab}(\mathcal{F}')$ that simultaneously accepts every positive example and rejects every negative example,

$$\exists \Delta \in \mathrm{Stab}(\mathcal{F}') \quad \big[\forall e \in E^+,\ \mathcal{F}' \models_{\Delta} e \ \wedge\ \forall e \in E^-,\ \mathcal{F}' \not\models_{\Delta} e\big].$$

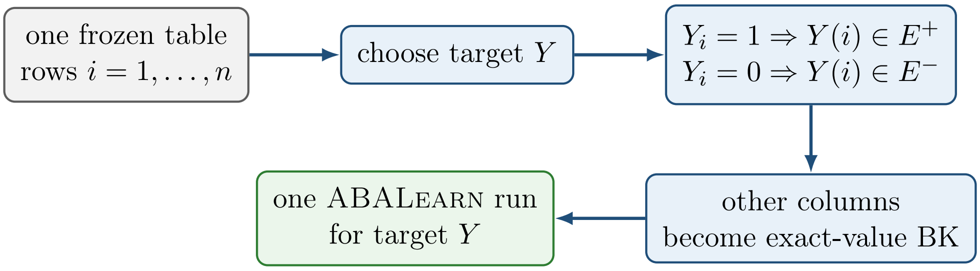

Figure 2: Target-wise encoding. All targets use the same rows; the selected target determines the positive and negative examples and the remaining columns determine the BK.

Cautious learning requires every positive example to be a cautious consequence and every negative example to be excluded as a cautious consequence,

$$\forall e \in E^+,\ \mathcal{F}' \models_{\text{caut}} e, \qquad \forall e \in E^-,\ \mathcal{F}' \not\models_{\text{caut}} e.$$

Thus the two modes quantify differently over $\text{Stab}(\mathcal{F}')$: brave learning uses one witnessing extension, while cautious learning uses acceptance in every stable extension.

2.6 Greedy and non-deterministic folding

Folding is controlled by `folding_mode`, together with companion options for selection and folding space. The two modes used here are as follows.

`folding_mode(greedy)`. A rote rule is folded in one deterministic pass against matching background predicates, under `folding_selection(mgr)` and `folding_space(bk)`. This is the folding regime of the AAMAS configuration.

`folding_mode(nd)`. Folding proceeds incrementally under a folding-token budget (`folding_steps`), with `folding_selection(any)` and `folding_space(all)`. This is the folding regime of the ECAI configuration and of the cautious nd configurations used later.

Each published configuration also sets learning semantics and `asm_intro`. When two named configurations are compared as wholes, several options change together.

2.7 Assumption introduction: `relto` and `sechk`

Assumption introduction is the transformation that makes an over-general folded rule defeasible by placing an assumption in its body and learning rules for the contrary of that assumption. In ABALEARN it is controlled by the option `asm_intro`, which takes one of two values: `relto` or `sechk`. Both are used after a folded candidate fails the configured post-folding entailment check. They differ in how the first assumption-repair step is chosen.

`asm_intro(relto)`. The learner first searches the current framework for an existing assumption that is relative to the body of the folded rule under consideration. If such an assumption is found, it is reused in the repaired rule. If none is found, the learner may generate a new assumption and continue along the subsequent generalisation path.

`asm_intro(sechk)`. The learner first generates a new provisional assumption for the folded rule and then subjects the resulting framework to the semantic-checking branch of generalisation (`gen5`/`gen6`), which inspects stable-extension consequences before deciding how the assumption is retained or revised.

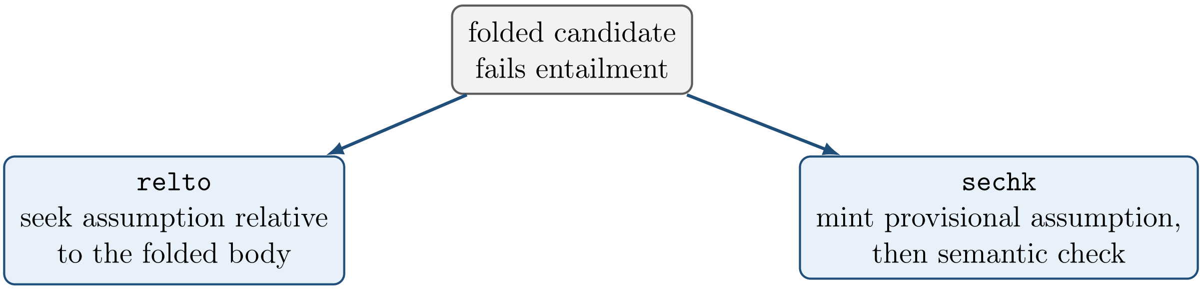

Figure 3: The two `asm_intro` options as alternative first steps of assumption introduction.

3 Causal fixtures and evaluator ground truth

This section records the generating processes, population certificates, evaluator-only mechanism references, and frozen samples used by the probes below. Mathematical labels are A, B, C, D; learner-safe identifiers are `a`,`b`,`c`,`d`. In emitted Prolog rules the capital argument (for example the `A` in `c(A)`) is the row identifier, not the causal variable A.

3.1 Principal fixture: binary AND collider (H0)

The principal fixture is the unshielded collider

$$G_0: \qquad A \longrightarrow C \longleftarrow B$$

with no A–B edge. The roots are mutually independent,

$$A \sim \text{Bernoulli}(4/5), \qquad B \sim \text{Bernoulli}(7/10), \qquad A \perp\!\!\!\perp B,$$

and the child is the deterministic AND mechanism

$$C = A \wedge B.$$

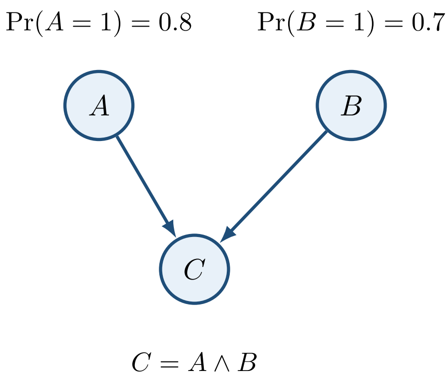

Figure 4: Principal generating DAG (H0). Randomness enters through the independent roots; the non-root mechanism is deterministic.

The induced observational population has four positive-probability atoms:

A	B	C	$P^*(A,B,C)$
0	0	0	$3/50 = 0.06$
0	1	0	$7/50 = 0.14$
1	0	0	$6/25 = 0.24$
1	1	1	$14/25 = 0.56$

The remaining four binary assignments $(0,0,1)$, $(0,1,1)$, $(1,0,1)$, and $(1,1,0)$ are structural zeros.

Causal sufficiency is declared by fixture design: the root exogenous noise terms are mutually independent and there is no omitted common cause. The causal Markov condition holds by construction of the joint,

$$P^*(a,b,c) = P^*(a)P^*(b)P^*(c \mid a,b).$$

Ordinary faithfulness was verified exactly by auditing every singleton-pair conditional-independence query against d-separation. The sole population independence is $A \perp\!\!\!\perp B$; conditioning on the collider yields $A \not\perp\!\!\!\perp B \mid C$, and each parent is dependent on C.

Under ordinary conditional-independence semantics, the skeleton is $A - C - B$ and the unshielded collider $A \to C \leftarrow B$ is oriented in the CPDAG; the corresponding MEC contains only G_0. This is the ordinary CI-based reading. Deterministic mechanisms can impose additional functional constraints beyond ordinary CI; no extended recovery object is asserted here. The graph, certificates, and mechanism reference are evaluator-only and are withheld from ABALearn.

3.2 Evaluator-only mechanism reference

For the deterministic child C, the canonical positive-state exact-value rule is

```
c(A) :- a_val_1(A), b_val_1(A).
```

On this fixture the formal truth-table rule, the population-supported rule, and the rule supported by the frozen sample coincide on that string. That string is the evaluator reference against which learned theories for target `c` are compared. Comparison uses rule bodies, assumptions, contraries, and the target's generating role together: a mechanism-aligned string is one form of evidence; an assumption-rich but sample-adequate theory is another object under the same reference.

The roots A and B are generated by independent stochastic mechanisms $A \sim \text{Bernoulli}(4/5)$ and $B \sim \text{Bernoulli}(7/10)$. Neither root is a deterministic function of the other observed variables, so when `a` or `b` is the learning target no underlying deterministic rule from those predictors should be learned.

3.3 Frozen sample for the principal fixture

The principal table is the target-free IID sample with $n = 30$ and seed 42:

Assignment	Count
$(1,1,1)$	17
$(1,0,0)$	7
$(0,1,0)$	4
$(0,0,0)$	2

All four population atoms appear. Under the binary exact-value encoding the implied example counts are

$$a : 24^+/6^-, \qquad b : 21^+/9^-, \qquad c : 17^+/13^-.$$

For child target `c`, every positive example has $(A, B) = (1, 1)$. For root target `a`, the predictor cell $(B, C) = (0, 0)$ contains both labels:

Row	A (target)	B	C	Class for target `a`
9	1	0	0	positive
26	0	0	0	negative

Rows 9 and 26 therefore share the same BK literals `b_val_0` and `c_val_0` while carrying opposite labels for A. No deterministic rule over (B, C) separates them. Target `b` has the symmetric ambiguity on predictor cells involving (A, C).

3.4 Controlled extensions: trailing and leading distractors (H2, H3)

$C = A \wedge B$ isolated

target-`c` BK order: $[a, b, d]$

$C = B \wedge D$

target-`c` BK order: $[a, b, d]$

Figure 5: Controlled four-variable extensions. Dashed red nodes are independent isolated variables. Serialisation order induces the stated BK predictor order for target `c`.

Trailing isolated D (H2). This fixture preserves the H0 collider, the H0 root laws $A \sim \text{Bernoulli}(4/5)$ and $B \sim \text{Bernoulli}(7/10)$, and the mechanism $C = A \wedge B$. It adds an independent isolated root $D \sim \text{Bernoulli}(1/2)$ with no edges. Variable order places `d` last, so for target `c` the exact-value BK predictors appear as $[a, b, d]$: the distractor is trailing. The evaluator rule for `c` remains

```
c(A) :- a_val_1(A), b_val_1(A).
```

The variables A, B, and the isolated D are each generated stochastically. When any of them is the learning target, there is no underlying deterministic rule from the remaining observed variables that should be learned. The frozen table used below is again an IID sample with $n = 30$ and seed 42 on this four-variable process.

BK-leading isolated A (H3). This fixture uses independent roots

$$A \sim \text{Bernoulli}(1/2), \qquad B \sim \text{Bernoulli}(4/5), \qquad D \sim \text{Bernoulli}(7/10),$$

with A isolated (no edges) and deterministic child $C = B \wedge D$. Variable order places `a` first, so for target `c` the BK predictors appear as $[a, b, d]$: the distractor is leading. The evaluator rule for `c` is

```
c(A) :- b_val_1(A), d_val_1(A).
```

The isolated variable A and the roots B and D are each generated stochastically. When any of them is the learning target, there is no underlying deterministic rule from the remaining observed variables that should be learned. The frozen table used below is an IID sample with $n = 30$ and seed 42 on this process.

Both extensions have verified Markov factorisation and exact ordinary faithfulness certificates. As with the principal fixture, the generating graph, certificates, and mechanism references are withheld from the learner.

4 Experimental design and comparison logic

This section records the learner configurations used in the probes, how strongly each designed contrast supports attribution to a single change, and the ordered probe map. Detailed outcomes appear in the experimental sequence that follows.

4.1 Learner configurations

Identity	Semantics	Folding	Sel.	Space	Steps	`asm_intro`
`aamas2025`	brave	greedy	mgr	BK	—	`relto`
`ecai2024`	brave	nd	any	all	10	`relto`
`baseline_cautious`	cautious	nd	any	all	10	`relto`
`greedy_cautious`	cautious	greedy	mgr	BK	—	`relto`
`nd_brave_sechk`	brave	nd	any	all	10	`sechk`
`nd_cautious_sechk`	cautious	nd	any	all	10	`sechk`

Here "Steps" is `folding_steps`, which bounds the non-deterministic folding-token budget; it is unused under greedy folding. All six identities also enable the post-learning checked-ASP artefact (`check_ic`). The cautious nd identities additionally pin `post_folding_test_entailment(true)`, which is likewise the effective default under the brave arms used here.

`aamas2025` and `ecai2024` are the repository presets corresponding to the AAMAS 2025 and ECAI 2024 learning-option bundles used in this work. `baseline_cautious`, `greedy_cautious`, `nd_brave_sechk`, and `nd_cautious_sechk` are experimental identities for controlled comparison. In particular, `greedy_cautious` matches `aamas2025` except for `learning_mode(cautious)`; it is an experimental cautious Greedy preset, not a published AAMAS method.

The H6a arms match `baseline_cautious` except for `folding_steps`, with $M \in \{1, 2, 5, 10\}$. H7a matches `ecai2024` except `asm_intro(sechk)`. H7b matches `baseline_cautious` except `asm_intro(sechk)`.

4.2 Strength of comparisons

Each probe is read at one of three attribution strengths.

- **Exact paired option ablation.** Fixture, sample, encoding, and the remaining learner options are held fixed; one learning option changes. Probes H4 and H4b change `learning_mode` under otherwise fixed nd / `relto` settings; H5 changes `learning_mode` under the fixed Greedy / `mgr` / BK / `relto` bundle; H6a changes `folding_steps`; H7a and H7b change only `asm_intro`.

- **Controlled data or fixture intervention.** The learner configuration is held fixed (or compared only within one arm); one designed property of the sample or fixture changes. H1 uses a nested seed-42 prefix $n = 25$ of the principal table under AAMAS; H2 adds a trailing isolated D under AAMAS; H3 uses the BK-leading isolated-A fixture under ECAI (with AAMAS as secondary); H6b varies nested sample size $n \in \{30, 60, 90\}$ at fixed `folding_steps(2)` under `baseline_cautious`.
- **Bundle-level contrast.** Several learner settings change together, so a behavioural difference is attributed to the configuration bundle rather than to one option. H0 compares `aamas2025` with `ecai2024` on the principal table. Where both exist, contrasts between `greedy_cautious` and `baseline_cautious` are likewise bundle-level (Greedy versus nd), as used secondarily in H5.

The strength level states how far a difference among outcomes, deltas, or traces can be attributed to the designed intervention of that probe.

4.3 Probe sequence

Probe	Question	Controlled intervention	What is inspected
H0	What do the Greedy and nd bundles learn for each target on the principal AND collider?	Principal fixture, frozen $n = 30$ seed 42; `aamas2025` and `ecai2024`; targets `a`,`b`,`c`	Exit status, deltas, and traces by target and arm.
H1	How does omitting the rare $(0, 0, 0)$ atom affect AAMAS root learning?	Nested seed-42 prefix $n = 25$ versus $n = 30$; AAMAS; same stream	Root deltas and the presence or absence of opposite-label witness rows.
H2	What does Greedy do with an irrelevant trailing covariate on the child?	Principal AND collider plus isolated D; AAMAS; target `c`	Whether `d` appears in the learned bodies relative to the D-free evaluator rule.
H3	What does nd learning do when an isolated covariate is first in BK?	BD AND collider with BK-leading isolated A; order $[a, b, d]$; ECAI primary on target `c`	First fold, retained BK literals, and the final child theory.
H4	How does switching brave nd to cautious nd affect root targets on the principal table?	Learning mode only, under fixed nd / `relto`; targets `a`,`b` with control `c`	Root exit status and the control child delta.
H4b	How does the same brave-to-cautious switch affect the child repair on the BK-leading fixture?	Learning mode only on H3 target `c`; nd / `relto` fixed	Shared structure versus the contrary-rule close under `alpha_3`.
H5	How does brave-to-cautious behave inside the Greedy bundle?	`learning_mode` only under Greedy / `mgr` / BK / `relto`; 18 paired cells across the completed AAMAS collections	Paired outcomes, deltas, traces, and runtime; secondary contrast to nd cautious where available.
H6a	How does the folding-token ceiling affect cautious-nd search cost on roots?	`folding_steps` $M \in \{1, 2, 5, 10\}$; principal table $n = 30$; `baseline_cautious`	Trace length, runtime, token bands, and control child behaviour.

Probe	Question	Controlled intervention	What is inspected
H6b	How does grounded sample size affect the same cautious-nd search?	Nested $n \in \{30, 60, 90\}$ at fixed $M = 2$; principal fixture; `baseline_cautious`	Search topology and per-band growth with n.
H7a	How do `relto` and `sechk` differ under brave nd?	`asm_intro` only; principal and BK-leading fixtures; seven target pairs	Paired deltas and the assumption-introduction control-flow fork.
H7b	How do `relto` and `sechk` differ under cautious nd?	`asm_intro` only; same two fixtures; `baseline_cautious` versus `nd_cautious_sechk`	Child deltas, root search cost, and whether runs complete within the wall-time limit.

5 Experimental sequence: H0–H7b

5.1 H0: baseline on the principal AND collider

Question and purpose. On the principal fixture $A \to C \leftarrow B$ with $C = A \wedge B$, frozen sample $n = 30$ seed 42, and exact-value encoding, what ABA frameworks do the Greedy bundle (`aamas2025`) and the nd bundle (`ecai2024`) emit when each of `a`, `b`, and `c` is the learning target?

Controlled comparison. The fixture, table, encoding, and target-specific examples were held fixed. The two arms differ as full configuration bundles: `aamas2025` uses brave learning with Greedy / `mgr` / BK folding and `relto`; `ecai2024` uses brave learning with nd / `any` / `all` folding and `relto`. The contrast is therefore bundle-level.

Arm	Target	Outcome	Emphasised result
AAMAS	`c`	`solved`	Assumption-free rule `c(A) :- a_val_1(A), b_val_1(A).`, string-identical to the evaluator mechanism.
AAMAS	`a,b`	`completed_no_solution`	Both-zero co-occurrence candidates cover opposite-label rows; contrary repair dead-ends under `relto`.
ECAI	`c`	`solved`	First-BK under-fold to `a_val_1`, repaired by an assumption attacked when `b_val_0`.
ECAI	`a,b`	`solved`	Brave per-row assumption construction on the ambiguous predictor cell; A (resp. B) remains underdetermined by the other observed variables.

Child target `c`

Emphasised observation. Under AAMAS, every positive example for `c` has $(A, B) = (1, 1)$. Greedy folding with BK space therefore builds the single maximal co-occurrence body $\{\texttt{a\_val\_1}, \texttt{b\_val\_1}\}$. After mgr deduplication the emitted delta is

```
c(A) :- a_val_1(A), b_val_1(A).
```

Brave entailment succeeds immediately: no negative example shares that body, so no assumption is introduced. The string coincides with the evaluator mechanism reference for C.

Under ECAI, nd folding takes one BK predicate at a time. From an early positive, the first fold replaces the example identifier by `a_val_1`, the first predictor in BK order, yielding the unary rule `c(A) :- a_val_1(A)`. That body also holds of every negative with $A = 1, B = 0$. Assumption introduction produces

```
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).
assumption(alpha_1(A)).
contrary(alpha_1(A),c_alpha_1(A)) :- assumption(alpha_1(A)).
```

Trace-supported explanation. The ECAI theory is sample-adequate under brave checking: c is derived when $a = 1$, unless the assumption is attacked when $b = 0$. Across the target rule and the contrary it cites both parents of C. Its ABA shape is nevertheless different from the AAMAS conjunction: the first fold latched onto the leading BK predictor, and the second parent enters only through contrary learning.

Root targets `a` and `b`

Emphasised observation. For target `a`, positives fall into two predictor cells, $(B, C) = (1, 1)$ and $(B, C) = (0, 0)$. The cell $(B, C) = (0, 0)$ also contains negatives: row 9 is positive with joint values $(1, 0, 0)$, while rows 26 and 30 are negative with $(0, 0, 0)$, yet all share the BK literals `b_val_0` and `c_val_0`. Target `b` is the symmetric case with the roles of A and B swapped.

Under AAMAS, greedy folding proposes the two Horn candidates

```
a(A) :- b_val_1(A), c_val_1(A).
a(A) :- b_val_0(A), c_val_0(A).
```

The both-one rule is accepted. The both-zero rule covers the negative rows 26 and 30, so brave entailment fails. The learner introduces an assumption on that body; contrary rote learning then cites those negatives, and greedy folding of the contrary reconstructs the same both-zero body. Under `asm_intro(relto)` a further relative assumption cannot be added for that body, the engine reports KO, and the cell ends as `completed_no_solution` with no emitted delta. Target `b` mirrors this path.

Under ECAI, nd folding again proceeds one predicate at a time. The learner first builds a safe block for the both-one positives,

```
a(A) :- alpha_1(A), b_val_1(A).
c_alpha_1(A) :- c_val_0(A).
```

That block covers positives with $(B, C) = (1, 1)$ and excludes negatives with $B = 1$ and $C = 0$. The both-zero positives remain uncovered, so a second ordinary target rule is still required.

Folding an early both-zero positive (row 8) produces the candidate `a(A) :- b_val_0(A)`. Brave entailment fails on that candidate. The same BK literals also hold of the negative examples in rows 26 and 30. The learner therefore introduces an assumption α_2 on the body, giving `a(A) :- alpha_2(A), b_val_0(A)`. Contrary rote learning cites rows 26 and 30 as grounds for `c_alpha_2`.

Folding that contrary first tries `b_val_0`. Under `relto` this returns KO against the already present α_2. The next fold tries `c_val_0`, yielding `c_alpha_2(A) :- c_val_0(A)`. Even that unary contrary remains too strong for the joint example set. The learner therefore introduces a further assumption α_3 on the contrary. Rote learning of `c_alpha_3` then cites the both-zero positives, including row 9. Folding that contrary re-uses α_2 together with `b_val_0`. The trace records this close as `found:`

`alpha_2/1 ... OK`. The final delta for target `a` is

```
a(A) :- alpha_1(A), b_val_1(A).
a(A) :- alpha_2(A), b_val_0(A).
c_alpha_1(A) :- c_val_0(A).
c_alpha_2(A) :- alpha_3(A), c_val_0(A).
c_alpha_3(A) :- alpha_2(A), b_val_0(A).
assumption(alpha_1(A)).
assumption(alpha_2(A)).
assumption(alpha_3(A)).
contrary(alpha_1(A),c_alpha_1(A)) :- assumption(alpha_1(A)).
contrary(alpha_2(A),c_alpha_2(A)) :- assumption(alpha_2(A)).
contrary(alpha_3(A),c_alpha_3(A)) :- assumption(alpha_3(A)).
```

Target `b` is the mirror obtained by swapping the roles of A and B.

Trace-supported explanation. The decisive geometry is the predictor cell $(B, C) = (0, 0)$. Row 9 is a positive example for target `a`. Its joint values are $(A, B, C) = (1, 0, 0)$. Row 26 is a negative example. Its joint values are $(0, 0, 0)$. Both rows carry the BK facts `b_val_0` and `c_val_0`. Any Horn rule that uses only those predictors therefore treats the two rows alike. It would derive `a` at both identifiers, or at neither.

The emitted α_2/α_3 nest works because assumptions are grounded per row identifier. At a fixed row i, the relevant instances are $\alpha_2(i)$ and $\alpha_3(i)$, with contraries `c_alpha_2`(i) and `c_alpha_3`(i). The local rules are

```
a(i) :- alpha_2(i), b_val_0(i).
c_alpha_2(i) :- alpha_3(i), c_val_0(i).
c_alpha_3(i) :- alpha_2(i), b_val_0(i).
```

On every both-zero row, both `b_val_0(i)` and `c_val_0(i)` hold. Accepting $\alpha_2(i)$ therefore derives `c_alpha_3`(i) and attacks $\alpha_3(i)$. Accepting $\alpha_3(i)$ derives `c_alpha_2`(i) and attacks $\alpha_2(i)$. The two assumptions cannot both be accepted at the same row.

This leaves two locally stable choices at each both-zero identifier. If $\alpha_2(i)$ is accepted and $\alpha_3(i)$ is rejected, then `a(i)` is derived. If $\alpha_3(i)$ is accepted and $\alpha_2(i)$ is rejected, then `a(i)` is blocked.

Brave learning asks for one witnessing stable extension Δ of the whole grounded framework. That extension must cover every positive example and exclude every negative example at once. Such a witness exists. On row 9 it accepts $\alpha_2(9)$ and rejects $\alpha_3(9)$. The rule `a(9) :- alpha_2(9), b_val_0(9)` then derives the positive example $(1, 0, 0)$. On row 26 it accepts $\alpha_3(26)$ and rejects $\alpha_2(26)$. The same BK facts are present, but without $\alpha_2(26)$ the rule for `a` does not fire, so the negative example $(0, 0, 0)$ is excluded. The two rows are therefore separated by their ground assumption atoms, even though their BK predictors are identical.

Row	Joint (A, B, C)	Choice in Δ	Effect
9	$(1, 0, 0)$ (positive)	$\alpha_2(9)$ in, $\alpha_3(9)$ out	`a(9)` derived
26	$(0, 0, 0)$ (negative)	$\alpha_2(26)$ out, $\alpha_3(26)$ in	`a(26)` blocked

Other stable extensions may reverse those local choices. Brave entailment requires only that at least one joint witness exists. The framework therefore passes brave checking on the finite sample.

For this mechanism, target `a` should have no learned solution: A is generated stochastically, and there is no underlying deterministic rule from the other observed variables. A solution was nevertheless found. Brave entailment permits one stable extension to assign opposite statuses to the ground assumptions $\alpha_2(i)$ and $\alpha_3(i)$ at different row identifiers. That per-row split covers the positive example $(1,0,0)$ and excludes the negative example $(0,0,0)$ even though the two rows share the same BK predictors. The ECAI `solved` outcome records that brave covering. It does not record recovery of a deterministic rule for A.

Evidential boundary. H0 is the establishing bundle contrast on one complete-support table. Procedural `solved` covers both the AAMAS mechanism-string match for `c` and the ECAI assumption-rich theories for `c` and the roots. Attribution of the arm difference is to the full AAMAS versus ECAI option bundle.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/learning_analysis.md`

5.2 H1: nested prefix omitting the rare $(0,0,0)$ atom

Question and purpose. On the principal AND collider under the fixed AAMAS bundle, what happens to root learning when the frozen seed-42 table is shortened to its first 25 rows, thereby omitting the only sample atoms $(0,0,0)$?

Controlled comparison. The fixture, population, seed sequence, AAMAS configuration, encoding, and target-wise protocol were held fixed. The sole designed change is the sample: H0 uses the first 30 rows of the seed-42 stream, while H1 uses the exact nested prefix of the first 25 rows. The shorter table is therefore a controlled support ablation of the longer one.

The H1 prefix has joint counts $(1,1,1) : 14$, $(1,0,0) : 7$, and $(0,1,0) : 4$. It contains no $(0,0,0)$ row. Those omitted rows are CSV rows 26 and 30 of the H0 table. The population still assigns mass $P^*(0,0,0) = 3/50 = 0.06$ to that atom. Only the finite sample omits it.

Under the binary exact-value encoding, the H1 example counts are $a : 21^+/4^-$, $b : 18^+/7^-$, and $c : 14^+/11^-$.

Target	H0 ($n = 30$)	H1 ($n = 25$)	Evaluator reading
`a`	`completed_no_solution`	`solved`, two Horn rules	H0 matches the intended root reading. H1 is a spurious finite-sample solve.
`b`	`completed_no_solution`	`solved`, mirror Horn rules	Same reading as for `a`.
`c`	`solved`, AND string	`solved`, AND string	Control unchanged in kind.

Emphasised observation. On the H1 table, AAMAS emits the root deltas

```
a(A) :- b_val_1(A), c_val_1(A).
a(A) :- b_val_0(A), c_val_0(A).
```

and

```
b(A) :- a_val_1(A), c_val_1(A).
b(A) :- a_val_0(A), c_val_0(A).
```

with zero assumptions. The child control remains

```
c(A) :- a_val_1(A), b_val_1(A).
```

The both-zero root rules are false as population implications. They hold on the H1 sample only because that sample contains no opposite-label witness in the relevant predictor cell.

Trace-supported explanation. The H0 and H1 AAMAS traces for target `a` follow the same greedy route up to the both-zero candidate. In both runs, greedy folding first accepts the safe rule

```
a(A) :- b_val_1(A), c_val_1(A).
```

It then proposes the both-zero body

```
a(A) :- b_val_0(A), c_val_0(A).
```

from an early both-zero positive (row 8 on both traces).

On the H1 prefix, brave entailment accepts that both-zero body immediately. No assumption is introduced. The folding queue empties. The run terminates as `solved`. The reason is support geometry: every remaining H1 row with `b_val_0` and `c_val_0` is a positive example for `a`. There is no negative example in that cell.

On the H0 table, the same both-zero body fails brave entailment. Rows 26 and 30 are negatives that share those BK literals. Assumption introduction and contrary learning then reconstruct the same both-zero body for the contrary. Under `relto`, a further relative assumption cannot be added. The engine reports KO and ends as `completed_no_solution`.

The learner configuration is unchanged. The population mechanism is unchanged. What flips the root outcome is the absence versus presence of a label conflict inside the greedy both-zero candidate cell. That missing conflict is exactly the rare joint $(0, 0, 0)$.

For these stochastic roots, the intended reading is that no deterministic rule from the other observed variables should be learned. The H0 AAMAS root outcome `completed_no_solution` matches that reading. The H1 AAMAS root `solved` outcome is a spurious finite-sample covering created by omitting the rare blocking atom. Target `b` mirrors the same pattern. Child `c` continues to emit the evaluator AND string on both prefixes and serves only as a control.

Evidential boundary. H1 is a nested-prefix support ablation under fixed AAMAS settings. It shows that greedy maximal co-occurrence search on this fixture is sensitive to whether a rare opposite-label witness appears in the table. The two prefixes are linked draws from one seed stream.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h1_support_ablation.md`

5.3 H2: add an irrelevant trailing covariate

Question and purpose. H2 asked whether Greedy would omit a variable which is statistically and causally irrelevant to the deterministic child once both of its values occur among positive examples.

Controlled comparison. The H0 collider and AND mechanism were preserved. An independent isolated binary root D was appended after A, B in target-c BK order. The main run used $n = 30$, seed 42, under AAMAS. Among the positive c rows, both $D = 1$ and $D = 0$ occur nine times.

Emphasised observation. AAMAS solves target c, but emits two D-specific rules:

```
c(A) :- a_val_1(A), b_val_1(A), d_val_1(A).
c(A) :- a_val_1(A), b_val_1(A), d_val_0(A).
```

The pair covers the sample and jointly behaves like AND across the observed values of D. It is nevertheless non-minimal and fails to coincide with the D-free evaluator mechanism.

Trace-supported explanation. Greedy BK folding exhausts matching exact-value BK literals within each positive co-occurrence body. A $D = 1$ row therefore retains `d_val_1`; a $D = 0$ row retains `d_val_0`. Subsumption can remove a specialised rule only when a suitable more general rule already exists. There is no later cross-rule anti-unification or literal-deletion phase that combines the complementary rules into:

```
c(A) :- a_val_1(A), b_val_1(A).
```

Evidential boundary. The secondary $n = 2$ prefix was recorded only as supporting context. H2's scientific signal is the balanced $n = 30$ distractor split. One fixture does not prove retention for every possible encoding or Greedy problem.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h2_irrelevant_covariate.md`

5.4 H3: put an irrelevant covariate first in BK

Question and purpose. H3 asked how nd learning behaves when an isolated variable precedes the two mechanism parents in target-c BK order. It was designed to expose whether the first folding opportunity becomes structural scaffolding for the learned theory.

Controlled comparison. H3 uses $B \to C \leftarrow D$, $C = B \wedge D$, and an isolated root A. The target-c order is $[a, b, d]$. The ECAI brave-nd run is primary; an AAMAS run on the identical table supplies a bundle-level contrast.

Emphasised observation. The ECAI trace begins:

```
folding result: c(A) <- [a_val_1(A)]
```

and the final ordinary target rules retain the isolated variable as two outer gates:

```
c(A) :- alpha_1(A), a_val_1(A).
c(A) :- alpha_2(A), a_val_0(A).
```

Later contrary rules cite b and d, but the emitted delta does not coincide with:

```
c(A) :- b_val_1(A), d_val_1(A).
```

On the same sample, AAMAS emits:

```
c(A) :- a_val_1(A), b_val_1(A), d_val_1(A).
c(A) :- a_val_0(A), b_val_1(A), d_val_1(A).
```

Thus both learner bundles solve c while unnecessarily organising the theory around isolated A.

Trace-supported explanation. Nd starts with one folding token and `any` selection. In the serialized BK, an a-value literal is the first admissible fold, so the first generalisation is based on A rather than the mechanism parents. When the unary fold over-generalises, the learner introduces an assumption and learns contraries around that rule. Later predicates can repair the theory, but the initial a-gate remains in the ordinary target rule. Greedy reaches a different but related failure through maximal co-occurrence: it keeps every matching literal and splits the AND rule by the two values of A.

Evidential boundary. H3 establishes a representation-order effect on this fixture. It does not show that the first BK variable must appear in every possible nd solution, or that any particular statistical ordering would be causally correct.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h3_bk_leading_distractor.md`

5.5 H4: change brave nd to cautious nd

5.5.1 H4: roots on the H0 table

Question and purpose. H4 returned to the H0 row-9/row-26 ambiguity and asked whether the brave root solution survives when the learning consequence relation is cautious.

Controlled comparison. The H0 fixture, $n = 30$ table, target-specific examples, BK, nd folding, `any` selection, `all` space, `relto`, and folding ceiling were fixed. The only learning-relevant change was `learning_mode(brave)` to `learning_mode(cautious)`. The comparator is the locked ECAI brave collection; the cautious arm is named `baseline_cautious`, not ECAI.

Target	Brave nd/`relto`	Cautious nd/`relto`
a	solved	no solution
b	solved	no solution
c	solved	solved, byte-identical delta

Emphasised observation. For root a, the traces coincide through the early $\alpha_1, \alpha_2, \alpha_3$ construction. They diverge at the attempted closure:

```
c_alpha_3(A) :- alpha_2(A), b_val_0(A).
```

The brave trace records the α_2 reuse as OK and completes the theory. The cautious trace records the same reuse as KO, exhausts the configured token budget, and emits no delta. Target b mirrors this. Control target c still solves with a byte-identical delta.

Trace-supported explanation. The brave construction can exploit different grounded assumption choices for the conflicting row identifiers inside one witnessing stable extension. Cautious checking asks whether the required consequences survive every stable extension. The decisive residual reuse does not, so the root gadget is rejected. The unchanged child control shows that cautious semantics has not simply prohibited assumption-bearing theories; it blocks this particular ambiguous construction.

Evidential boundary. The cautious root result is relative to the tested configuration and ceiling $M = 10$. It does not prove that no cautious theory exists under all search strategies or budgets, and no-solution is not root-mechanism recovery.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h4_cautious_vs_brave.md`

5.5.2 H4b: child c on the H3 table

Question and purpose. H4b asked a narrower structural question: given H3's leading-a gate, how does cautious checking change the later b, d split when learning child c?

Controlled comparison. H3 fixture, sample, target c, BK order, nd settings, and `relto` were fixed. Only brave versus cautious learning changed.

Emphasised observation. Both arms solve. The leading-a gates and the α_1-nested versus α_2-flat contrary skeleton are shared. Exactly one closing rule changes:

```
% Brave:
c_alpha_3(A) :- alpha_1(A), a_val_1(A).

% Cautious:
c_alpha_3(A) :- d_val_1(A).
```

Trace-supported explanation. Cautious checking rejects reuse of α_1 on the $a = 1$ side, so search continues until `d_val_1` is accepted. The experiment therefore locates the semantic effect in the α_3 closure. It does not attribute the leading-a gates to cautious learning: those gates were already present in the H3 brave theory.

Evidential boundary. Neither solution recovers the direct B, D AND rule; both retain isolated A. H4b concerns the internal split and repair, not whether A disappears.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h4b_cautious_split_under_a.md`

5.6 H5: duplicate all Greedy runs under cautious learning

Question and purpose. H5 asked whether the brave-to-cautious change that mattered under nd would alter any completed Greedy result or trace, and whether Greedy-cautious no-solution searches incur the long replays seen under cautious nd.

Controlled comparison. Five completed deterministic Bucket 3 AAMAS collections were duplicated under experimental `greedy_cautious`: H0 $n = 30$, H1 $n = 25$, H2 $n = 30$, H2c $n = 2$, and H3 $n = 30$. This gave 18 exact target-wise pairs. `folding_mode(greedy)`, `mgr`, BK space, `relto`, data, BK, and examples were fixed; only the learning mode changed.

Emphasised observation. Across all 18 pairs:

- every outcome matches;

- every solved delta is byte-identical;
- the traces are isomorphic apart from entailment-logging surfaces;
- no new assumption or contrary structure appears; and
- cautious runtime is approximately 1.6–2.3× brave runtime, with mean ratio about 1.84, while remaining below one second.

The invariance includes the desirable H0/H1 child conjunctions, the H1 spurious root rules, the H2/H3 distractor-retaining child rules, and the H0 root no-solutions.

Trace-supported explanation. The inspected Greedy paths either accept strict Horn rules or fail before an assumption-bearing solution is produced. Consequently, changing how stable extensions are quantified does not change the emitted deltas on these cells. Greedy also terminates without the nd folding-token replay loop, explaining why its cautious root no-solutions remain short.

Evidential boundary. H5 establishes semantic inertness only across this completed 18-pair set. The large difference between Greedy-cautious and baseline cautious nd is a search-bundle difference, not an isolated consequence of `folding_mode`.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h5_greedy_cautious.md`

5.7 H6: explain cautious-nd search cost

5.7.1 H6a: folding-token ceiling

Question and purpose. H6a asked why H4's cautious root no-solution traces contained roughly 5,000 lines and took about one minute, and whether the configured `folding_steps(10)` ceiling was driving repeated work.

Controlled comparison. The H0 $n = 30$ sample and `baseline_cautious` bundle were fixed. `folding_steps` varied over $M \in \{1, 2, 5, 10\}$; the $M = 10$ collection was the existing H4 run and was not regenerated.

M	Target	Outcome	Runtime (s)	Trace lines	New assumptions
1	a	no solution	6.638	549	9
2	a	no solution	13.117	1047	18
5	a	no solution	32.775	2541	45
10	a	no solution	69.952	5031	90

Target b mirrors this. Target c solves at every ceiling within `tokens(1)`, with the same 134-line trace, approximately 1.3-second runtime, and byte-identical delta.

Emphasised observation. For the no-solution roots, trace length and runtime grow approximately linearly with M across the tested values. Each permitted token band adds about nine new assumptions and roughly 500 trace lines without producing a different accepted theory.

Trace-supported explanation. `folding_steps(M)` acts as a token-budget ceiling, not as a literal count of successful folds. The decisive root KO already occurs in the first band. When budget remains, the nd controller increments the token allowance and replays essentially the same failed assumption-introduction motif:

$$\text{lines}(M) \approx C_{\text{fixed}} + M\,C_{\text{failed band}}.$$

Through $M = 10$, the additional budget buys repeated failed search rather than a new explanation.

Evidential boundary. The experiment does not show that $M = 1$ is always sufficient. Other learnable tasks may require more tokens. It explains the tested root cost but does not yet provide a general stopping rule.

5.7.2 H6b: nested sample size

Question and purpose. H6b asked how cautious-nd trace size changes when the folding-token topology is held fixed but more grounded IID rows are supplied.

Controlled comparison. The H0 population, seed 42, `baseline_cautious` bundle, and $M = 2$ were fixed. Nested prefixes used $n \in \{30, 60, 90\}$.

n	a lines	a runtime (s)	NEW	KO	c lines
30	1047	13.117	18	32	134
60	1641	22.169	18	32	200
90	2241	32.291	18	32	266

All four H0 population atoms are already present at $n = 30$. Their multiplicities, ordered as $(0, 0, 0), (0, 1, 0), (1, 0, 0), (1, 1, 1)$, are:

$$\begin{aligned} n &= 30 : (2, 4, 7, 17), \\ n &= 60 : (4, 10, 12, 34), \\ n &= 90 : (5, 17, 18, 50). \end{aligned}$$

Emphasised observation. The root search topology stays fixed at 18 new assumptions and 32 KOs, while target-a trace length increases by 594 and 600 lines for each additional 30 rows. Target b mirrors this. Target c retains the same delta but its trace grows by 66 lines per 30 rows.

Trace-supported explanation. Additional rows create more grounded example identifiers, rote rules, entailment checks, subsumption work, and contrary batches inside each fixed search band. They do not introduce a new supported value pattern in this comparison. The approximately linear line growth therefore comes from per-row grounding and bookkeeping while the procedural topology remains unchanged.

Evidential boundary. H6b is a nested-n experiment, not a direct deduplication ablation. Multiplicities still carry empirical frequency information, and removing duplicate rows can alter row order and learner behaviour. The result is local to $n = 30, 60, 90$, not a universal complexity law.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h6_folding_and_n_ablation.md`

5.8 H7: change assumption introduction from `relto` to `sechk`

5.8.1 H7a: brave nd

Question and purpose. H7a asked whether the learned rules and trace path change when the assumption-introduction option is changed inside the brave nd bundle. It also asked whether any difference follows a consistent relationship to serialized BK order and causal role.

Controlled comparison. Within each pair, H0 or H3 fixture, $n = 30$ sample, target, examples, BK, brave semantics, nd folding, `any` selection, `all` space, folding ceiling, and other settings were fixed. The locked `ecai2024`/`relto` collection was compared with experimental `nd_brave_sechk`/`sechk`. There were seven target-wise pairs.

The target roles must be kept explicit:

Cell	Causal role	Evaluative use
H0/H3 c	deterministic child	desirable mechanism target
H0 a	parent of c, root as target	root diagnostic
H3 b	parent of c, root as target	role-peer of H0 a
H3 a	isolated root	leading-distractor diagnostic
H0 b, H3 d	other parent roots	supporting mirrors

Emphasised observation. Both arms solve all seven cells, but every paired `delta.aba` differs:

Cell	BK order	Outcome	`sechk` variables	Additional `relto` variables
H0 a	b, c	solved / solved	b	c
H0 b	a, c	solved / solved	a	c
H0 c	a, b	solved / solved	a	b
H3 a	b, c, d	solved / solved	b	c, d
H3 b	a, c, d	solved / solved	a	c, d
H3 c	a, b, d	solved / solved	a	b, d
H3 d	a, b, c	solved / solved	a	b, c

For H0 child c:

```
% relto
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).

% sechk
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- alpha_2(A), a_val_1(A).
c_alpha_2(A) :- alpha_1(A), a_val_1(A).
```

Both theories use the first a-predicate as the outer gate. `relto` later reaches mechanism parent b, whereas `sechk` closes an assumption chain around a. On H3 child c, `relto` reaches b, d, while `sechk` uses only the first, isolated variable a.

Trace-supported explanation. After the failed post-fold entailment check, `relto` first searches for a relative assumption. A KO can return control to folding and admit later BK literals. `sechk` instead generates a provisional assumption and enters `gen5`/`gen6`. Under brave semantics on these cells, the resulting assumption chain closes before the later folds reached by `relto`.

Every H7a `sechk` delta therefore cites only its first BK predictor block. This is not always variable a: for target a, the first predictor is b. Byte-identical H0/H3 `sechk` root deltas consequently reflect a shared learner template, not a shared causal role.

Evidential boundary. H7a does not establish that `sechk` is worse than `relto`, or that first-BK closure is universal. It records a bounded brave-nd interaction on seven exact input-matched pairs.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h7a_relto_vs_sechk.md`

5.8.2 H7b: cautious nd

Question and purpose. H7b asked whether the H7a `relto`/`sechk` difference transfers unchanged when both arms use cautious rather than brave learning.

Controlled comparison. H0 and H3 fixtures, frozen $n = 30$ tables, targets, examples, BK, nd folding, `any` selection, `all` space, and folding ceiling were fixed. Locked `baseline_cautious`/`relto` was compared with experimental `nd_cautious_sechk`/`sechk`. The comparator was not ECAI: both arms used cautious semantics.

Seven targets were run, but the evidence has unequal strength:

Question	Cell	Outcome pair	Evidential use
Q1	H0 c	solved / solved	Comparable deltas and traces.
Q2	H3 c	solved / solved	Comparable deltas and traces.
Q3	H0 b	no solution / no solution	Comparable failed-search cost.
Q4	H3 b	timeout / timeout	Non-comparable: both logs empty.
Q5	H3 a	timeout / timeout	Non-comparable: both logs empty.

Emphasised observation. Both cautious policies solve child c on both tables, but their paired deltas differ. For H0 c:

```
% baseline_cautious / relto
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).

% nd_cautious_sechk / sechk
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- alpha_2(A), a_val_1(A).
c_alpha_2(A) :- b_val_1(A).
```

`sechk` still inserts a new assumption layer around the first a-predicate, but cautious checking prevents the brave early closure and search eventually reaches b.

For H3 c, both policies retain the isolated-a gates and cite mechanism parents b, d somewhere in their contrary structures. The `sechk` theory is thicker:

Arm	Assumptions	Trace lines	Trace character
relto	3	257	Relative KO then a lean path to b, d.
sechk	5	899	Deeper nests and repeated NEW/gen5/KO cycles.

For H0 root b, neither cautious arm emits a delta. The sechk trace has approximately 13,818 lines, compared with 5,118 for relto, because it makes many more provisional-assumption and gen5 attempts. H3 targets a and b time out in both arms with empty stdout files; those matched timeout labels reveal no trace-level similarity.

Trace-supported explanation. The same procedural fork as H7a occurs after failed post-fold entailment, but cautious semantics changes which provisional repairs survive. Relative reuse often receives a cautious KO and can expose another fold. The sechk path first adds a provisional assumption; when cautious checking rejects an early closure, further nesting and backtracking can eventually reach later b or d predicates. Thus H7b retains the H7a assumption-policy intervention while demonstrating that its observable effect is conditioned by brave versus cautious entailment.

Evidential boundary. The child theories remain non-canonical and, on H3, distractor-retaining. Mechanism-parent occurrence in a contrary is not mechanism recovery. The failed-root comparison concerns cost, and the empty-log H3 timeouts are non-comparable.

Primary evidence record: docs/experiments/qualitative/M13-C3-binary-collider-and/h7b_cautious_relto_vs_sechk.md

6 Findings

6.1 Finding 1: Greedy solutions retain all represented BK predictors

6.2 Finding 2: nd solutions are organised around the first BK predictor

6.3 Finding 3: brave nd can solve predictor-ambiguous root targets

6.4 Finding 4: cautious-nd failed-search cost grows with the folding-token ceiling

6.5 Finding 5: cautious-nd trace size grows with grounded sample size

6.6 Finding 6: relto and sechk produce different assumption-introduction paths

7 Implications